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$$H_0: p_0(x) = \begin{cases} 1, & x \in (0,1) \\ 0, & x \notin (0,1) \end{cases}$$

$$H_1: p_1(x) = \begin{cases} \frac{e^{1-x}}{e-1}, & x \in (0,1) \\ 0, & x \notin (0,1) \end{cases}$$

a) $n=1$

$$L = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \{0,1\} \geq c$$

$$1-x \geq \ln c(e-1)$$

$$x \leq 1 - \ln c(e-1) = A$$

$$G_{\text{opt}}: x \leq A; \quad P(x \leq A | H_0) = \alpha$$

$$\int_0^A p_0(x) dx = \alpha \Rightarrow A = \alpha$$

$$G_{\text{opt}}: x \leq \alpha$$

$$\underline{\alpha_1 = \alpha}$$

$$\begin{aligned} W_{\alpha} &= P(x \leq A | H_1) = \int_0^{\alpha} p_1(x) dx = \\ &= \int_0^{\alpha} \frac{e^{1-x}}{e-1} dx = -\frac{e^{1-x}}{e-1} \Big|_0^{\alpha} = \underline{(1-e^{-\alpha}) \frac{e}{e-1}} \end{aligned}$$

$$\alpha_2 = 1 - (1 - e^{-d}) \frac{e}{e-1}$$

b) $n = 2$

$$L = \frac{L_1}{L_0} = \frac{p_1(x_1) \cdot p_2(x_2)}{p_0(x_1) \cdot p_0(x_2)} = \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2} \geq C$$

$$e^{-x_2-x_1} \geq \frac{C(e-1)^2}{e^2}$$

$$-x_2 - x_1 \geq \ln\left(\frac{C(e-1)^2}{e^2}\right)$$

$$G_{\text{Krit}}: x_1 + x_2 \leq A$$

$$P(x_1 + x_2 \leq A | H_0) = d$$

$$\iint_{x_1+x_2 \leq A} 1 \cdot dx_1 \cdot dx_2 = \frac{1}{2} A^2 = d \Rightarrow A = \sqrt{2d}$$

$$G_{\text{Krit}}: x_1 + x_2 \leq \sqrt{2d}, \quad d_1 = d$$

$$\begin{aligned} W &= P(x_1 + x_2 \leq A | H_1) = \iint_{x_1+x_2 \leq A} \frac{e^2}{(e-1)^2} e^{-x_1-x_2} dx_1 dx_2 = \\ &= \frac{e^2}{(e-1)^2} \int_0^A dx_1 \int_0^{A-x_1} e^{-x_2-x_1} dx_2 = \frac{e^2}{(e-1)^2} (1 - e^{-A} - e^{-A}) \end{aligned}$$

$$\alpha_2 = 1 - W = 1 - \frac{e^2}{(e-1)^2} (1 - e^{-A} - e^{-A})$$

c) $L = \frac{L_1}{L_0} = \prod_{i=1}^n \frac{p_1(x_i)}{p_0(x_i)} \geq C$

$$\ln L = \sum_{i=1}^n \underbrace{\ln\left(\frac{p_1}{p_0}\right)}_{\eta_i} \geq \ln C$$

$$\eta_i = \ln\left(\frac{p_i}{p_0}\right) = \ln\left(\frac{e^{1-x_i}}{e-1}\right) = \ln \frac{e}{e-1} - x_i$$

$$\frac{\sum_{i=1}^n \eta_i - n\mu_{\eta}}{\sqrt{nD_{\eta}}} \rightsquigarrow N(0,1)$$

$$P(\ln L \geq \ln C | H_0) = \alpha$$

$$\ln L = n \ln \frac{e}{e-1} - \sum_{i=1}^n x_i \geq \ln C$$

$$G_{\eta}: \sum_{i=1}^n x_i \leq A$$

$$P\left(\frac{\sum_{i=1}^n x_i - n\mu_x}{\sqrt{nD_x}} \leq \frac{A - n\mu_x}{\sqrt{nD_x}} | H_0\right) = \alpha$$

$$\text{ges } H_0: \mu_x = \frac{1}{2}$$

$$D_x = \frac{1}{12}$$

$$\frac{A - \frac{n}{2}}{\sqrt{\frac{n}{12}}} = u_{\alpha}$$

$$A = \frac{n}{2} + u_{\alpha} \sqrt{\frac{n}{12}}$$

$$C_{\eta}: \sum_{i=1}^n x_i \leq \frac{n}{2} + u_{\alpha} \sqrt{\frac{n}{12}}$$

$$\alpha_1 = \alpha$$

$$W = P\left(\sum_{i=1}^n x_i \leq A | H_1\right) =$$

$$= P\left(\frac{\sum_{i=1}^n x_i - n\mu_x}{\sqrt{nD_x}} \leq \frac{A - n\mu_x}{\sqrt{nD_x}} | H_1\right)$$

$$\text{ges } H_1: \mu_x = \int_0^1 \frac{e^{1-x}}{e-1} x dx = \frac{e-2}{e-1}$$

$$\mu_{x^2} = \int_0^1 \frac{e^{1-x}}{e-1} x^2 dx = \frac{2e-5}{e-1}$$

$$D_x = \frac{2e-5}{e-1} - \left(\frac{e-2}{e-1}\right)^2 = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$W = \int_{-\infty}^{\beta} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \rightarrow 1, n \rightarrow \infty, m, n$$

$$\beta = \frac{\frac{n}{2} + \sqrt{\frac{n}{12}} - n \frac{e-2}{e-1}}{\sqrt{n \frac{e^2-3e+1}{(e-1)^2}}} \sim \sqrt{n} \left(\frac{1}{2} - \frac{e-2}{e-1} \right) \rightarrow \infty, n \rightarrow \infty$$

$\Rightarrow$ критическое значение.

d) $G_{\text{кр}}: x_{\min} < C \quad P(x_{\min} < C | H_0) = \alpha$

$$H_0: \xi \sim R(0, 1)$$

$$H_1: x_{\min} \sim 1 - (1 - F(x))^n$$

$$1 - (1 - F(C))^n = \alpha$$

$$(1 - F(C))^n = 1 - \alpha$$

$$F(C) = 1 - \sqrt[n]{1 - \alpha} = C$$

$$G_{\text{кр}}: x_{\min} < 1 - \sqrt[n]{1 - \alpha} \quad \alpha_1 = \alpha$$

$$W = P(x_{\min} < C | H_1)$$

$$F_1(x) = \int_{-\infty}^x p_1(t) dt = \int_0^x \frac{e^{1-t}}{e-1} dt = \frac{e}{e-1} (1 - e^{-x}),$$

$$W = 1 - (1 - F(C))^n = 1 - \left(1 + \frac{e^{1-x}}{e-1} - \frac{e}{e-1} \right)^n \Big|_{x=C} =$$

$$= 1 - \left(1 + \frac{e^{\sqrt[n]{1-\alpha}}}{e-1} - \frac{e}{e-1} \right)^n$$

$$\alpha_2 = \left(1 + \frac{e^{\sqrt[n]{1-\alpha}}}{e-1} - \frac{e}{e-1} \right)^n$$

$$e^{\sqrt[n]{1-\alpha}} = \exp\left(\exp\left(\frac{1}{n} \ln(1-\alpha)\right)\right) = \exp\left(1 + \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right)\right) =$$

$$= e^1 \left(1 + \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right) \right)$$

$$L_2 = \left(1 + \frac{e}{e-1} \left(1 + \frac{1}{n} \ln(1-d)\right) + O\left(\frac{1}{n}\right) - \frac{e}{e-1}\right)^n =$$

$$= \left(1 + \frac{e}{e-1} \cdot \frac{1}{n} \ln(1-d) - \underbrace{\frac{e}{e-1} O\left(\frac{1}{n}\right)}_{\rightarrow 0, n \rightarrow \infty}\right)^n = ?$$

$$= ? \quad L_2 \rightarrow e^{\frac{e}{e-1} \ln(1-d)} = (1-d)^{\frac{e}{e-1}} \neq 0, \quad n \rightarrow \infty \Rightarrow$$

$\Rightarrow$ предел найдем.